OO-L-90B
August 4, 1969
SUPERSEDING
Fed. Spec. OO-L-90A
September 14, 1962

FEDERAL SPECIFICATION

LATHE, BRAKE DRUM

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one type of floor-mounted electrically powered lathe, with equipment and accessories, for truing brake drums of commercial and military series automotive vehicles and for truing brake discs when specified.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

CC-M-636 - Motor, Alternating-Current (Fractional Horsepower).
CC-M-641 - Motor, Alternating-Current (Integral Horsepower, 200 HP and Smaller).
FF-B-171 - Bearings, Ball, Annular (General Purpose).
FF-R-185 - Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning.
FF-B-187 - Bearing, Roller, Tapered.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26 15:52
Check the source to verify that this is the current version before use.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-T-152 - Treatment, Moisture and Fungus-Resistant, of Communications, Electronic and Associated Electrical Equipment.
MIL-M-18058 - Machinery, Metal and Wood-Working, Support Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-I-24092 - Insulating Varnish, Electrical, Impregnating.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U. S. Military Property.
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MS-19081 - Bearings, Roller, Tapered Single Row of Rollers, Normal Angle.

Army Drawings:

7025887 - Drum Assembly, Wheel Brake.
7328372 - Brake Drum and Back Assembly.

(Copies of Military Specifications, Standards, Drawings, and Publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:58Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS), Handbook:

H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

Tire and Rim Association, Inc.:

Tire and Rim Data Book With Military Supplement.

(Application for copies should be addressed to the Tire and Rim Association, Inc., Comand Bldg., 34 N. Hawkins Ave., Akron, Ohio 44313.)

National Electrical Manufacturers Association (NEMA) Standards:

NEMA Standards.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th St., New York, N. Y. 10017.)

American Gear Manufacturers Association (AGMA) Standards:

201.02 - Tooth Proportions for Coarse-Pitch Involute Spur Gears.

390.02 - Gear Classification Manual.

(Application for copies should be addressed to the American Gear Manufacturers Association, One Thomas Circle, Washington, D. C. 20005.)

United States of America Standards Institute (USASI):

B5.17 - Markings for Identifying Grinding Wheels and Other Bonded Abrasives.

B7.1 - American Standard Safety Code for the Use, Care, and Protection of Abrasive Wheels.

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th St., New York, N. Y. 10016.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:58Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the supplier shall furnish one complete lathe for first article inspection. The first article may be either a pre-production model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The brake drum lathe shall include all the features, components, and accessories necessary for truing automotive brake drums as specified herein. When specified (see 6.2), the lathe shall include all the features, components, and accessories necessary for truing automotive brake discs for all American made trucks and passenger cars. The lathe shall be new and of the latest design characteristics capable of meeting the performance requirements specified herein. The lathe shall be electric motor driven, with automatic and hand feeds. The equipment shall be convenient to operate from the front of the machine, and the operations such as feed selection, tool bar positioning, grinder-quill positioning, movement of all slides, and main spindle traverse shall be within easy reach of the operator. All parts subject to wear, breakage, or distortion shall be accessible for replacement, adjustment, and repair.

3.2.1 Interchangeability. All replaceable parts shall be manufactured to such standards as will permit replacement without the need for modification.

3.2.2 Standard parts. Military standard parts or commercial standard parts shall be used provided they conform to the requirements of this specification, except as otherwise specified herein.

3.2.3 Fastening devices. All screws, pins, bolts, and similar fastening devices shall be installed with means for preventing loss of tightness and adjustment. All such parts, when subject to removal or adjustment, shall not be swaged, peened, staked, or otherwise permanently deformed.

3.2.3.1 Screw threads. Screw threads shall be in inch units of measurement and comply with NBS H28.

3.2.4 Gears. All gears and pinions shall be fabricated in the inch system and shall conform to AGMA 390.02. Spur gear tooth form shall conform to AGMA 201.02.

3.2.5 Bearings. Bearings shall be sleeve type or antifriction type. When other than tapered roller bearings are used, suitable thrust bearings shall be provided. Antifriction bearings, when furnished, shall be interchangeable with, or be in accordance with, the requirements of FF-B-171, FF-B-185, or FF-B-187, as applicable. Sleeve-type bearings shall be bronze or babbitt. All bearings shall be protected from dust and chips.

3.2.6 Lubrication. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provided with means for lubrication.

3.2.7 Safety devices. Each machine shall be furnished with safety devices. Parts which are hazardous to the operator shall be guarded. Protection against electric shock shall be provided. When specified (see 6.2), safety devices shall be provided as required by the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:58Z
Check the source to verify that this is the current version before use.

3.2.8 Surfaces. External surfaces shall be cleaned and smooth, and all edges shall be rounded or beveled. All bearing surfaces between matching parts shall be finished by machining, grinding, or scraping to necessary tolerances to provide the required accuracy. All castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free of sand, dirt, fins, sprues, scale, flux, and other harmful materials.

3.2.9 Materials. Materials shall be as specified herein. Materials not specifically designated shall be of good commercial quality.

3.2.9.1 Corrosion resistance. All exposed metallic materials shall be inherently corrosion resistant or treated to be corrosion resistant.

3.2.10 Plates. All instruction plates and indicating dials shall be permanently marked in bold type.

3.3 Construction. Construction shall be free from any characteristics or defects that may render the lathe unsuitable or inefficient for the intended purpose.

3.3.1 Castings and forgings. All castings and forgings shall be free from blowholes, porosity, hard spots, cracks, scale, inclosures, cold shuts, mismatching, and other defects that may affect the structural strength of the castings or forgings. Strength and other physical and chemical properties of the castings and forgings shall be adequate to meet the performance requirements specified herein. Castings and forgings shall not be subjected to any reclaiming process without the authorization of the procuring activity.

3.3.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be performed in accordance with commercial practice.

3.3.3 Electrical connection. Connections of conductors and terminal parts shall be of the screw, pressure, or solder type. When soldered connections are used, the conductors and terminal parts shall be mechanically secured before soldering. Resin base fluxes only shall be used in soldering operations. Connections to screw type terminals shall be mechanically secured with a means to prevent loss of tightness.

3.4 Performance and product characteristics. The lathe shall be capable of truing commercial and military automotive brake drums within the sizes and vehicle classes specified herein with one initial machine setup, for each drum. The main drive motor shall provide sufficient power to meet the performance requirements specified. The lathe shall be capable of turning all brake drums 17-1/4 inches diameter or less while mounted as single or dual assemblies not exceeding 1,300 pounds or 35 inches maximum overall width (7.00-16 through 14.00-24 tire sizes as listed in the Tire and Rim Data Book with Military Supplement) without removal of the wheel, tires, or tubes from the brake drum and hub. It must be capable of turning larger sizes within specified size range with the wheels removed. The lathe shall be capable of turning towed vehicle brake drums within the sizes specified in 3.4.1.

3.4.1 Minimum physical characteristics. Unless otherwise specified (see 6.2), minimum physical characteristics of the brake drum lathe shall be as follows:

(a) Drum diameter range capacity - 8.9 to 22 inches.
(b) Drum cutting depth - 8 inches.
(c) Number of speed changes - 3.
(d) Cutting speed range - 30 to 100 revolutions per minute (r.p.m.).
(e) Swing - 60 inches.
(f) Feed range - 0.005-0.016 inch per revolution.
(g) Number of feeds - 2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:58Z
Check the source to verify that this is the current version before use.

3.4.2 Machine accuracy. The brake drum lathe and accessories shall not exceed the following tolerances:

(a) Spindle taper run-out (concentricity) 0.0005 inch total indicator reading (t.i.r.).
(b) Arbor run-out (concentricity) 0.001 inch t.i.r. at any point along its length up to 16 inches from the spindle nose.
(c) Adapters (short radii cones, taper cones, sleeves, bushings, and spacers) run-out and concentricity to their axis 0.001 inch t.i.r.
(d) Carriage alignment 0.002 inch t.i.r. with the axis of the spindle, at any point up to and including the necessary travel to true an 8-inch face depth drum.

3.4.3 Performance accuracy. The brake drum lathe shall be capable of truing all brake drums specified herein within the following tolerances:

(a) Concentricity (turned surface to axis of hub). The turned surface of the brake drum shall be concentric with the axis of the bearing surfaces in the hub within 0.002 inch t.i.r.
(b) Taper. The turned surface of the brake drum shall be true with the axis of the bearing surfaces of the hub within 0.002 inch t.i.r. per 8 inches.
(c) Finished surface. The final turned surface finish of the brake drum shall not have a finish numerically greater than 100 microinches arithmetical average (a.a.).

3.4.4 Performance accuracy for brake discs. When the requirement for truing brake discs is specified in 3.2, the lathe shall be capable of truing both friction surfaces of brake discs within the tolerances specified below:

(a) Each finish trued surface of the brake disc shall not deviate from the brake disc true plane in excess of $\pm$ 0.0005 inch.
(b) Each finish trued surface of the brake disc shall be parallel to its opposite counterpart surface with a total tolerance not to exceed 0.001 inch.
(c) The final trued surface finish of the brake disc shall not have a finish numerically greater than 30 microinches a.a. (It may be necessary to furnish a grinder with each machine to produce the above required finishes.)

Note: Brake disc true plane - the true plane of the brake disc is a plane, perpendicular to the axis of the functional bearing cup surfaces.

3.5 Detail of components. The lathe shall consist of, but not be limited to, the following principal component parts.

3.5.1 Base. The base shall be of cast iron or steel fabrication. Beds shall be seasoned or stress relieved. Steel-weldment structures shall be stress relieved. The base shall be of sufficient strength to insure rigidity, and insure permanency of position when operating under load.

3.5.1.1 Ways. All ways shall be finished to provide an accurate, uniform bearing surface for the matching ways. All V-way and flat-way construction shall be adequate to maintain the required accuracy.

3.5.2 Carriage. The carriage, when the lathe is not a sliding spindle type, shall slide on ways with means provided for adjustment to compensate for wear.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:58Z
Check the source to verify that this is the current version before use.

3.5.2.1 Feed drive. The feed screw and feed rod may be exposed and the feed mechanism totally enclosed. A handwheel or clutch lever for engaging the automatic feed shall be provided. Feed changes shall be easily adjustable.

3.5.2.2 Automatic stops. An automatic stop device shall be provided which will shut off the feed at a predetermined point, preventing damage to the equipment for both grinding and turning.

3.5.3 Spindles and arbors. Spindles and arbors shall be high-grade alloy steel, properly heat treated and ground on all aligning surfaces. Spindles and arbors shall be of a size to insure the accuracy required of this type of lathe and so mounted as to permit turning brake drums of the size specified. The arbors shall be mounted in the spindle by means of a suitable taper with a positive lock and release system which will prevent damage to spindle or arbor. The length of the arbor shall be adequate to mount brake drums of all vehicles within the sizes specified.

3.5.3.1 Spindle drive. Spindle drive shall be through V-belts, chains, gears, or combinations thereof. Spindle speeds shall be variable through use of stepped pulleys, infinitely variable drive mechanisms, or quick-change-speed gears. A minimum of three spindle speed ranges shall be provided. Gear drives shall be totally enclosed. Easily adjustable tensioning devices and guards shall be provided where belt or chain drives are employed. All necessary motor mounting brackets, pulleys, belts, chains, sprockets, gears, countershafts, and other parts required to place the lathe in successful operation shall be furnished. Visual means shall be provided to show spindle speeds throughout the entire speed range specified herein.

3.5.4 Cross-slide. The cross-slide shall be equipped with a feed dial, graduated in increments to indicate 0.001-inch movements. The cross-slide shall have provisions for locking in any position and adjustment to compensate for wear. The cross-slide shall furnish provisions for mounting the tool bar rigidly and securely in its proper position on the cross-slide.

3.5.5 Tool bar. The lathe tool bar shall be made from steel and shall be designed to hold the cutting tool rigidly and securely, and prevent change in the rake angle of the cutting tool. When specified (see 6.2), the tool bar shall be designed so that it can be rotated 180° in the clamp for turning drums with external braking surfaces.

3.5.6 Electrical equipment.

3.5.6.1 Motors. The motors shall conform to the requirements of CC-M-636 and CC-M-641, as applicable. The lathe shall be furnished with a continuous duty, induction squirrel-cage type, minimum 1 horsepower motor with a temperature rise of not more than 50°C. above ambient temperature, thermal overload protected. Unless otherwise specified (see 6.2), the motor shall function on 115 volts, single phase, 60 hertz current. Mounting shall be in accordance with NEMA Standards. Motors shall be protected from dirt and grit and shall be accessible for lubrication and maintenance and shall be equipped with sealed-for-life, antifriction bearings. A start-stop switch shall be mounted and wired, convenient to the operator.

3.5.6.2 Electrical components. Electrical components of the lathe including starters, relays, switches, wiring, and their inclosures shall conform to and shall be located in accordance with NEMA Standards, except as specified herein. There shall be no component part used where the requirements exceed the electrical component manufacturer's rating for the part.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:58Z
Check the source to verify that this is the current version before use.

3.5.6.3 Light fixture. The lathe shall be provided with a work light located in a position which will not interfere with controls, yet will afford full illumination at the cutting tool area. The light shall be equipped with a separate switch that shall function independently of the motor switch.

3.5.7 Case. When specified (see 6.2), a suitable sturdy case of sheet metal or wood shall be furnished to contain the grinding attachment, gages, and all accessories for transportation of the equipment.

3.5.8 Grinding attachment. When specified (see 6.2), a grinding attachment shall be furnished with each lathe with six grinding wheels suitable for grinding cast-iron and hardened-steel brake drums. A suitable guard, an abrasive-wheel dresser having a mounted octahedron diamond of 1/4-carat minimum size, feed mechanism, brackets, and necessary equipment to make a complete unit shall be included. The grinding attachment shall be capable of driving the abrasive wheel at a working surface speed of 3,600 to 7,000 feet per minute.

3.5.8.1 Grinding wheels. The grinding wheels shall be marked, handled, mounted, and used in accordance with USASI B5.17 and USASI B7.1.

3.5.9 Crane attachment. When specified (see 6.2), a crane attachment (less chain falls) shall be provided for the lathe and shall be capable of being attached to the lathe in such a manner as to form an integral unit. The crane shall be of sufficient strength to handle all the drums with tires and wheels attached, within the capacities specified.

3.5.10 Accessories. Unless otherwise specified (see 6.2), all necessary accessories for truing brake drums of military vehicles shall be furnished as well as those accessories required for truing drums of all standard models of United States produced commercial trucks and passenger cars. For military vehicles, such accessories shall be provided to fit all bearing cups from the minimum size bearing cup MS 19081-58 through the maximum size bearing cup MS 19081-173. Floating drum adapters shall be provided for brake drum, Army Drawing 7328377 and brake drum, Army Drawing 7025887 as used on military vehicles M38A1 (1/4 ton, 4 by 4 utility truck) and M151 (1/4 ton, 4 by 4 utility truck), respectively. Turning of drums shall be accomplished without chatter.

3.5.10.1 Disc brakes. When specified (see 6.2), accessories required for turning disc type brakes shall be provided.

3.5.11 Spare parts and tools. The following spare parts and tools shall be furnished with each lathe:

One complete set of extra belts, when used.
One complete set of tools (other than common) necessary for operation of the equipment.
Two tool bits, carbide for steel and cast iron.
Twelve tool tips, replaceable carbide for steel and cast iron.

3.5.12 Outboard support. Unless otherwise specified (see 6.2), an outboard support on the brake drum lathe shall be provided. The outboard support shall be a portable type of equipment easily removed for loading and unloading of wheel assemblies from end of spindle. An adjustment device shall be provided to allow easy alignment adjustment of arbor(s) fully loaded with hub, drum, wheel and tire assemblies.

3.6 Electromagnetic interference control. When specified (see 6.2), electromagnetic interference control shall be in accordance with the requirements of MIL-STD-461 for class IIB equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:58Z
Check the source to verify that this is the current version before use.

3.7 Fungus control. When specified (see 6.2), electrical components and circuit elements shall be treated for fungus resistance in accordance with the type II requirements of MIL-T-152, except that the statement of treatment is not required. Windings of the electric motor shall be given two impregnating coats of varnish conforming to the requirements of MIL-I-24092, type M, class 155 through 220 (manufacturer's option).

3.8 Painting. Unless otherwise specified (see 6.2), painting and finishing of the brake drum lathe shall be in accordance with the supplier's standard practice provided the following minimum criteria are met or exceeded: all surfaces to be painted shall have been cleaned to be free of all foreign matter detrimental to painting, and at least one coat of primer and one coat of enamel have been applied.

3.9 Identification of product. The brake drum lathe shall be marked for identification in accordance with MIL-STD-130, including serial number.

3.9.1 Civil agencies. The brake drum lathe shall be marked in a legible and permanent manner with the serial number and manufacturer's name or trademark of known character so that the source of manufacturer can be readily identified.

3.10 Handbook of instructions. Unless otherwise specified (see 6.2), the supplier shall furnish two copies of a handbook of instructions with each brake drum lathe supplied under this specification. The handbook shall contain operating instructions, illustrations, parts list, and safety precautions to insure proper operation and maintenance of the lathe. Additional copies of the handbook and the mailing address to which the additional copies are to be sent shall be as specified by the procuring activity (see 6.2).

3.10.1 Handbook approval. When specified (see 6.2), a preliminary copy of the instruction handbook shall be submitted to the contracting officer for approval prior to publication and delivery.

3.11 Workmanship. Workmanship of the brake drum lathe shall be of the acceptable quality prevailing among manufacturers normally producing lathes of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:58Z
Check the source to verify that this is the current version before use.

4.2.1 First article inspection. First article inspection shall be applied to the preproduction model or initial production item specified in 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.3 and all tests under 4.4 and 4.5. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be applied to the brake drum lathes being submitted for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the following:

(a) Product examination (see 4.3).
(b) Machine accuracy tests (see 4.4.1 and 4.4.1.1).
(c) Preparation for delivery (see 4.6).

4.2.2.1 Inspection lot. For quality conformance inspection, a lot shall consist of all brake drum lathes from one manufacturer submitted for acceptance at one time.

4.2.2.2 Sampling. When applicable (see 6.2), sampling for lot acceptance shall be in accordance with MIL-STD-105 at acceptable quality levels (AQL) specified herein. Samples for the tests specified shall be selected at random from the item samples that passed the examination.

4.3 Examination. The lathe(s) shall be examined using the following classification of defects and AQL. Any lathe(s) found defective shall be rejected, and if the number of defective lathe(s) exceeds the AQL limit specified, the lot represented by these samples shall be rejected.

4.3.1 Classification of defects.

Major defects
(AQL shall be 1.5 percent defective)

101. Component parts that are not new; evidence of reclaimed, used, or salvaged components (see 3.2).
102. Parts not accessible for repair (see 3.2).
103. Equipment not convenient to operate (see 3.2).
104. Interchangeability and marking not as specified (see 3.2.1).
105. Standard parts not as specified (see 3.2.2).
106. Fastening devices not as specified (see 3.2.3).
107. Screw threads not as specified (see 3.2.3.1).
108. Gears not as specified (see 3.2.4).
109. Bearings not as specified (see 3.2.5).
110. Lubrication not as specified (see 3.2.6).
111. Safety devices not as specified (see 3.2.7).
112. Materials not as specified (see 3.2.9).
113. Corrosion resistance not as specified (see 3.2.9.1).
114. Construction not as specified (see 3.3).
115. Defective castings or forgings (see 3.3.1).
116. Defective welding, brazing, or soldering (see 3.3.2).
117. Electrical connections not as specified (see 3.3.3).
118. Product characteristics not as specified (see 3.4).
119. Brake drum size range capacities not as specified (see 3.4.1).
120. Base not as specified (see 3.5.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:58Z
Check the source to verify that this is the current version before use.

Major defects (cont'd.)

121. Ways not as specified (see 3.5.1.1).
122. Carriage not as specified (see 3.5.2).
123. Feed drive not as specified (see 3.5.2.1).
124. Automatic stops not as specified (see 3.5.2.2).
125. Spindles and arbors not as specified (see 3.5.3).
126. Spindle drive not as specified (see 3.5.3.1).
127. Cross-slide not as specified (see 3.5.4).
128. Tool bar not as specified (see 3.5.5).
129. Motors not as specified (see 3.5.6.1).
130. Electrical components not as specified (see 3.5.6.2).
131. Light fixture not as specified (see 3.5.6.3).
132. Case not as specified (see 3.5.7).
133. Grinding attachment not as specified (see 3.5.8).
134. Grinding wheels not as specified (see 3.5.8.1).
135. Crane attachment not as specified (see 3.5.9).
136. Accessories not as specified (see 3.5.10).
137. Spare parts and tools not as specified (see 3.5.11).
138. Outboard support not as specified (see 3.5.12).
139. Handbook of instructions not as specified (see 3.10).

Minor defects
(AQL shall be 6.5 percent defective)

201. Surfaces not as specified (see 3.2.8).
202. Electromagnetic interference control not as specified (see 3.6).
203. Fungus control not as specified (see 3.7).
204. Painting not as specified (see 3.8).
205. Identification of product not as specified (see 3.9).
206. Handbook approval not as specified (see 3.10.1).
207. Workmanship not as specified (see 3.11).

4.4 Test methods.

4.4.1 Machine accuracy tests. Each lathe selected for tests shall be tested to determine conformance with machine accuracy requirements specified in 3.4.2. Movable spindles shall be checked at maximum extended position.

4.4.1.1 Accuracy test on brake discs. When the requirement for truing brake discs is specified in 3.2, each lathe selected for tests shall also be tested to determine conformance with the accuracy requirements specified in 3.4.4.

4.4.2 Performance test. Brake drums with hub, tire, and wheel assemblies shall be set up and trued as a typical operation. The concentricity, taper, and surface roughness value shall be within the limits specified (see 3.4.3). The lathe shall be capable of truing one brake drum of the smallest diameter (8.9 inch minimum diameter), with wheel, tire (7.00-16), hub and drum attached; and one brake drum of the larger diameters (17.25 inch maximum diameter) with dual wheels, tires (14.00-24), hub and drum attached.

4.4.3 Spindle speed test. The lathe shall be tested with a tachometer to determine compliance with spindle speed range and number of speeds required. Failure to comply with the minimum speed range and the minimum speed changes specified shall be cause for rejection (see 3.4.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:58Z
Check the source to verify that this is the current version before use.

4.4.4 Endurance test. The lathe shall be operated for 200 consecutive start-stop cycles while rotating a 17.25 inch drum with dual wheels, tires (14.00-24), hub, and drum attached. The lathe shall be started, brought to a minimum spindle speed of 60 r.p.m. and then allowed to come to a complete stop before starting the next cycle. Upon completion of the 200 cycles, the machine shall be tested in accordance with 4.4.1. Failure to pass this test shall be cause for rejection. The spindle driving gears shall be visually inspected for overload or fatigue, breakage, cracking, scoring, or plastic yielding. Any evidence of such defects shall be cause for rejection.

4.5 Electromagnetic interference control test. Unless otherwise specified (see 6.2), equipment requiring electromagnetic interference control shall be tested in accordance with table II of MIL-STD-461 to determine compliance with the requirements of section 3 of this specification.

4.6 Preparation for delivery examination. Preservation, packaging, packing, and marking for shipment shall be examined to determine compliance with section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall conform to level A, B, or C requirements of MIL-M-18058, as specified (see 6.2).

5.2 Marking for shipment.

5.2.1 Military agencies. Interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.2.2 Civil agencies. In addition to any special marking specified in the contract or order, each unit and intermediate package and shipping container shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use. The brake drum lathes covered by this specification are intended for use in truing brake drums of the commercial and military type, and for truing brake discs when specified.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) First article, when required (see 3.1).
(c) Brake disc truing design, when required (see 3.2).
(d) Special safety devices, when required (see 3.2.7).
(e) Physical characteristics, if different(see 3.4.1).
(f) External tool bar, when required (see 3.5.5).
(g) Motor voltage, phase, and current, if different (see 3.5.6.1).
(h) Case, when required (see 3.5.7).
(i) Grinding attachment, when required (see 3.5.8).
(j) Crane attachment, when required (see 3.5.9).
(k) Accessories, if different (see 3.5.10).
(l) Disc brake accessories, when required (see 3.5.10.1).
(m) Outboard support, if different (see 3.5.12).
(n) Electromagnetic interference, when required (see 3.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:58Z
Check the source to verify that this is the current version before use.

(o) Fungus control, when required (see 3.7).
(p) Painting, if different (see 3.8).
(q) Handbook of instructions, if different, and address where additional copies are to be sent (see 3.10).
(r) Handbook approval, when required (see 3.10.1).
(s) First article inspection, if different (see 4.2.1).
(t) Quality conformance inspection, if different (see 4.2.2).
(u) Sampling, when required (see 4.2.2.2).
(v) Electromagnetic interference control test, if different (see 4.5).
(w) Selection of applicable level of preservation, packaging, and packing (see 5.1).

CUSTODIANS:
Army - WC
Navy - YD
Air Force - 82

Review activities:
Army - EL, SM
Navy - YD
Air Force - 82

User activity:
Navy - MC

Preparing activity:
Army - WC

CIVIL AGENCIES:
COM - NBS
GSA - FSS
PO - ORE
VA - DMS

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:58Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION
00-L-90B Lathe, Brake Drum

ORGANIZATION	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM APR 63 1426 REPLACES NAVSHIPS FORM 4863 WHICH IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:58Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
Headquarters
U.S. Army Weapons Command
Rock Island Arsenal
Rock Island, Illinois 61202
OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

Commanding General
Hq, U.S. Army Weapons Command
ATTN: AMSWE-RDT
Rock Island Arsenal
Rock Island, Ill. 61202

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:58Z
Check the source to verify that this is the current version before use.